

Roll Number: _____

Thapar Institute of Engineering & Technology
Department of Electronics and Communication Engineering

M.E. (II SEM): VLSI

Course Code: PVL333

Course Name: System On Chip

March 15, 2024, 3:00 PM

Time: 02 Hrs.

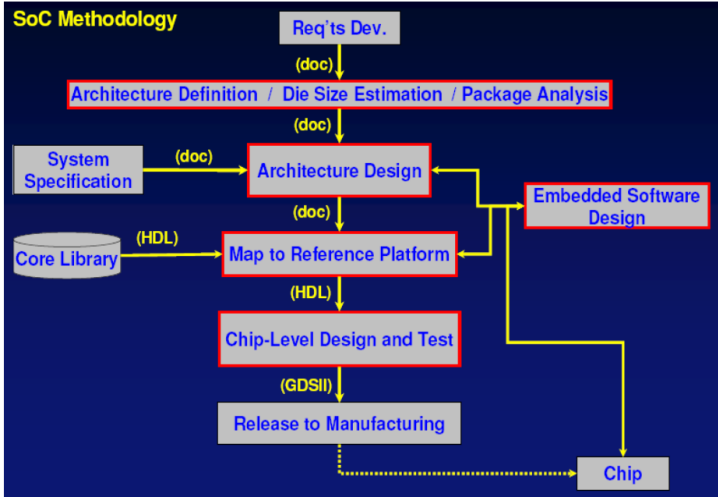
Max. Marks: 35.

Name of Faculty: Dr. Robin Singla

Note: Attempt all questions & assume missing data, if any, suitably

Q. No.	Questions	Marks	CO	BL
Q.1	Write brief note on the following. Draw the neat diagram wherever required. (a) ASIC design flow (b) EDA tools	(7)	C02	L2 & L3
Q.2	Discuss the top-down SOC design flow in detail.	(7)	C01	L2 & L3
Q.3	Differentiate between front-end and back-end SOC architecture design.	(7)	C03	L2 & L3
Q.4	Explain in detail the BUS based intra chip communication in SOC.	(7)	C04	L2 & L3
Q.5	What are the SOC design metrics? Which different techniques can be used for improve a specific design metrics?	(7)	C03	L2 & L3

Q.1	Write brief note on the following. Draw the neat diagram wherever required. (a) ASIC design flow (b) EDA tools	(7)																																																																											
	<pre> graph TD BM[Behavioral Model VHDL/Verilog] --> Synthesis GLN[Gate-Level Netlist] GLN --> DFT/BIST & ATPG GLN GLN --> Standard Cell IC & FPGA/CPLD TLN[Transistor-Level Netlist] TLN --> DRC & LVS Verification PL[Physical Layout Map/Place/Route] PL --> IC Mask Data/FPGA Configuration File PL BM <--> VF1[Verify Function] GLN <--> VF2[Verify Function] TLN <--> VFT[Verify Function & Timing] PL <--> VT[Verify Timing] </pre> The diagram illustrates the ASIC design flow. It starts with a Behavioral Model (VHDL/Verilog) which undergoes Synthesis to produce a Gate-Level Netlist. The Gate-Level Netlist is then processed by DFT/BIST & ATPG to generate Test vectors. The Gate-Level Netlist is also converted to a Transistor-Level Netlist using Standard Cell IC & FPGA/CPLD. The Transistor-Level Netlist undergoes DRC & LVS Verification to produce a Physical Layout (Map/Place/Route). The Physical Layout is then used to generate IC Mask Data/FPGA Configuration File. Verification steps include Verify Function for the Behavioral Model and Gate-Level Netlist, Verify Function & Timing for the Transistor-Level Netlist, and Verify Timing for the Physical Layout.																																																																												
	<table border="1"> <thead> <tr> <th>Purposes</th><th>Tool</th><th>company</th></tr> </thead> <tbody> <tr> <td rowspan="6">RTL Simulations</td><td>VCS</td><td>Synopsys</td></tr> <tr> <td>Xcelium</td><td>Cadence Design System</td></tr> <tr> <td>NC-Sim</td><td>Cadence Design System</td></tr> <tr> <td>QuestaSim</td><td>Mentor Graphics (Now Siemens)</td></tr> <tr> <td>ModelSim</td><td>Mentor Graphics (Now Siemens)</td></tr> <tr> <td>ISE Simulator</td><td>Xilinx</td></tr> <tr> <td rowspan="2">RTL Signoff</td><td>Incisive</td><td>Cadence Design System</td></tr> <tr> <td>SpyGlass</td><td>Synopsys</td></tr> <tr> <td rowspan="5">Logic Synthesis</td><td>JasperGold</td><td>Cadence Design System</td></tr> <tr> <td>Fusion Compiler (RTL-to-GDSII solution)</td><td>Synopsys</td></tr> <tr> <td>Design Compiler</td><td>Synopsys</td></tr> <tr> <td>Genus Synthesis Solution</td><td>Cadence Design System</td></tr> <tr> <td>RTL Compiler (upgraded to Genus)</td><td>Cadence Design System</td></tr> <tr> <td rowspan="2">LEC</td><td>Conformal</td><td>Cadence Design System</td></tr> <tr> <td>Formality</td><td>Synopsys</td></tr> <tr> <td rowspan="3">Place and Route</td><td>Innovus</td><td>Cadence Design System</td></tr> <tr> <td>IC Compiler (ICC)</td><td>Synopsys</td></tr> <tr> <td>Encounter (upgraded to Innovus)</td><td>Cadence Design System</td></tr> <tr> <td rowspan="2">RC Extraction</td><td>Quantus RC Extraction</td><td>Cadence Design System</td></tr> <tr> <td>StarRC</td><td>Synopsys</td></tr> <tr> <td rowspan="4">STA</td><td>primeTime</td><td>Synopsys</td></tr> <tr> <td>Tempus</td><td>Cadence Design System</td></tr> <tr> <td>Tweaker</td><td>Dorado (Now a part of Synopsys)</td></tr> <tr> <td>Encounter Timing System (ETS)</td><td>Cadence Design System</td></tr> <tr> <td rowspan="2">IR Analysis</td><td>Voltus</td><td>Cadence Design System</td></tr> <tr> <td>RedHawk</td><td>Ansys</td></tr> <tr> <td rowspan="4">Physical Verification (Fill, DRC, LVS, Antenna, PERC)</td><td>Calibre</td><td>Mentor Graphics (Now Siemens)</td></tr> <tr> <td>IC Validator</td><td>Synopsys</td></tr> <tr> <td>Pegasus</td><td>Cadence Design System</td></tr> <tr> <td>PVS (Physical Verification System)</td><td>Cadence Design System</td></tr> <tr> <td>Power simulation</td><td>Jules</td><td>Cadence Design System</td></tr> </tbody> </table>	Purposes	Tool	company	RTL Simulations	VCS	Synopsys	Xcelium	Cadence Design System	NC-Sim	Cadence Design System	QuestaSim	Mentor Graphics (Now Siemens)	ModelSim	Mentor Graphics (Now Siemens)	ISE Simulator	Xilinx	RTL Signoff	Incisive	Cadence Design System	SpyGlass	Synopsys	Logic Synthesis	JasperGold	Cadence Design System	Fusion Compiler (RTL-to-GDSII solution)	Synopsys	Design Compiler	Synopsys	Genus Synthesis Solution	Cadence Design System	RTL Compiler (upgraded to Genus)	Cadence Design System	LEC	Conformal	Cadence Design System	Formality	Synopsys	Place and Route	Innovus	Cadence Design System	IC Compiler (ICC)	Synopsys	Encounter (upgraded to Innovus)	Cadence Design System	RC Extraction	Quantus RC Extraction	Cadence Design System	StarRC	Synopsys	STA	primeTime	Synopsys	Tempus	Cadence Design System	Tweaker	Dorado (Now a part of Synopsys)	Encounter Timing System (ETS)	Cadence Design System	IR Analysis	Voltus	Cadence Design System	RedHawk	Ansys	Physical Verification (Fill, DRC, LVS, Antenna, PERC)	Calibre	Mentor Graphics (Now Siemens)	IC Validator	Synopsys	Pegasus	Cadence Design System	PVS (Physical Verification System)	Cadence Design System	Power simulation	Jules	Cadence Design System	
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	 SoC Methodology <pre> graph TD Req[Req'ts Dev.] -- "(doc)" --> ArchDef[Architecture Definition / Die Size Estimation / Package Analysis] ArchDef -- "(doc)" --> ArchDes[Architecture Design] SysSpec[System Specification] -- "(doc)" --> ArchDes CoreLib[(Core Library)] -- "(HDL)" --> MapRef[Map to Reference Platform] ArchDes -- "(doc)" --> MapRef MapRef -- "(HDL)" --> ChipLevel[Chip-Level Design and Test] ChipLevel -- "(GDSII)" --> Release[Release to Manufacturing] ArchDes --> ESD[Embedded Software Design] ESD --> Chip[Chip] Release -.-> Chip </pre>	
Q.3	Differentiate between front-end and back-end SOC architecture design.	(7)
	• ASIC Back End Design Flow consists of following steps: - ➤ Acquiring Gate Level Netlist and Design Constraints ➤ Partitioning ➤ Floorplaning ➤ Placement ➤ Routing ➤ Verification	
Q.4	Explain in detail the BUS based intra chip communication in SOC.	(7)
	BUS based communication Different structure Different type of BUS BUS Clocking BS Topologies	
Q.5	What are the SOC design metrics? Which different techniques can be used for improve a specific design metrics?	(7)
	Portability Timing Cost Functionality Reliability Performance Time to market reusability	
